USING CLINICAL REPORT FREE TEXT TO DETERMINE THE LEVEL OF DIFFICULTY OF NASOGASTRIC TUBE PLACEMENT FOR A COMPUTER VISION MODEL

by Group 2a

CHME0039: Artificial Intelligence in Healthcare Group Project

University College London

London

7 May 2026

Contents

1. Problem Statement: Background & Literature Review 3
2. Project Plan 5
3. Summary of Testing 7
4. Summary of Results 9
5. Conclusions & Key Lessons 12
References **13**

Table of Tables

Table 1: Summary of regression metrics 9

1. Problem Statement: Background & Literature Review

Computer vision (CV) models are increasingly integrated into clinical workflows to assist radiologists, with deep learning systems demonstrating diagnostic accuracy comparable to specialist clinicians across a range of radiological tasks (Liu et al., 2019). Despite this potential, integration into routine clinical workflows remains limited due to concerns around reliability, generalisability, and variability in performance (Javaid et al., 2024; Lindroth et al., 2024). Complex cases with poor image quality or subtle abnormalities yield uncertain predictions, posing challenges for safe clinical deployment.

Nasogastric tube (NGT) placement verification exemplifies this challenge, where misplacement can result in severe complications. Radiologist interpretation of CXRs is the gold standard for placement confirmation, yet radiologists are not always immediately available (Creeden et al., 2025). CV models show promise, with neural network ensembles achieving an AUC of 0.82 for identifying satisfactory NGT placement and improving junior clinicians' decision-making (Drozdov et al., 2023). Yet, performance declines in complex cases (Drozdov et al., 2023) where radiologists' inter-rater variability also exists (Creeden et al., 2025), suggesting that difficult cases challenge both CV models and clinical experts–with human-AI collaboration enabling better resource allocation and reducing misinterpretations.

Model performance also varies across datasets and clinical environments. Zech et al. (2018) demonstrated that models trained on one hospital system may fail to generalise effectively to others, indicating that classification "difficulty" is context-dependent. Such inconsistency highlights the critical need for more robust difficulty prediction methods. Given radiology workforce shortages and increasing demand (The Royal College of Radiologists, 2024), identifying difficult cases is essential for prioritising expert review while automating routine cases.

However, a practical challenge is that the "difficulty" of a CXR case is rarely labelled explicitly in datasets. X-ray images are commonly accompanied by free-text radiology reports (Johnson et al., 2019). Clinical natural language processing (NLP) research shows that texts are capable of predicting coding complexity of medical cases even before coding completion (Xu et al., 2023). Thus, reports may contain meaningful linguistic signals indicative of classification difficulty.

Therefore, this study investigates whether radiology reports contain linguistic signals predictive of CV model confidence in NGT placement detection. The study aims to develop a machine learning (ML) model predicting classification difficulty from report text, enabling prioritisation of difficult cases for expert review, and improving clinical workflow efficiency.

2. Project Plan

2.1 Overview

The project followed a multimodal three-stage ML pipeline implemented over 10 weeks within a Trusted Research Environment (TRE). The pipeline comprised:

1. CV model for confidence scores
2. NLP of radiology reports
3. Regression model predicting confidence from text embeddings

The pipeline aimed to develop a working binary classifier achieving at least 70% accuracy.

2.2 Conceptualisation

A challenge of this project was translating an abstract concept, classification difficulty, into a measurable, reproducible construct suitable for supervised learning. Difficulty was defined as the CV model's confidence in identifying NGT placement.

Binary difficulty labels were derived from the CV model's predicted confidence score using a fixed threshold:

- Cases where the model's confidence score was ≥ 0.80 were labelled "EASY", indicating sufficient model certainty in its NGT placement classification
- Cases where the model's confidence score was < 0.80 were labelled "HARD", indicating model uncertainty and potential need for expert clinical review

2.3 Data Pre-processing

CXR images were pre-processed for the ResNet-50 model by resizing to 224 x 224 pixels, normalising pixel values, and converting to tensors, after which the model generated confidence scores (0-1) for each image.

Radiology reports were encoded using ClinicalBERT to produce 768-dimensional embeddings, capturing domain-specific semantic content. Confidence scores and corresponding text embeddings were paired using index-based matching to form the final dataset used to train and evaluate the regression model.

2.4 CV Model

A custom ResNet-50 binary classifier was trained on pre-processed CXR images to perform NGT placement classification (properly placed vs. misplaced). ResNet-50 (He et al., 2016) was selected for its strong performance on medical image classification tasks and its suitability for transfer learning from ImageNet, reducing training time while leveraging established visual feature representations. The model provided confidence scores between 0 and 1 for each image, which served as the target variable for classification difficulty. Difficulty labels were assigned using a 0.80 threshold, as defined in section 2.2.

2.5 NLP Model

ClinicalBERT (Huang et al., 2019), pre-trained on over two million clinical notes, was used to encode free-text radiology reports into 768-dimensional contextual embeddings capturing domain-specific semantic content.

2.6 Evaluation & Analysis

Model outputs were evaluated against the generated difficulty labels. Regression metrics included Mean Absolute Error (MAE) and R-squared (R^2) to assess the accuracy and explanatory power of predicted confidence scores relative to actual CV model outputs. Binary classification performance was assessed using thresholded predictions, with class distribution examined to identify class imbalance.

3. Summary of Testing

3.1 Testing Objective

The objective of the study was to investigate whether radiology reports contain sufficient linguistic information to predict NGT placement-detection difficulty for CV models (see section 1).

3.2 Testing Framework

Our testing framework employed a dual-output regression with thresholding, where BERT embeddings from radiology reports were used to predict confidence scores (0-1), which were then classified into “EASY” or “HARD” using a threshold of 0.80.

Regression is appropriate because our target variable (confidence) was continuous. The dual-output design enables both quantitative assessment of predictive strength and categorical interpretation for practical assessment.

The 0.80 threshold was selected as it aligns with medical imaging literature where confidence ≥ 0.80 indicates sufficient model certainty for automated processing (Boehringer et al., 2023; Vermeulen et al., 2023).

3.3 Evaluation Models & Metrics

Two complementary models were trained and evaluated: linear regression as an interpretable baseline for detecting direct relationships between text embeddings and confidence scores, and random forest models to capture non-linear patterns through ensemble learning.

Regression performance was assessed using MAE to quantify prediction error and R^2 to evaluate explanatory power. Following thresholding at 0.80, classification performance was evaluated using F1-score and Recall. Balanced Accuracy accounted for class imbalance, and the Matthews Correlation Coefficient (MCC) was a robust overall measure incorporating all confusion matrix elements.

3.4 Validation Strategy

To ensure robust evaluation, 5-fold cross-validation was employed, with random partitioning (random_state = 42) to ensure reproducibility of results. Performance metrics were averaged across all folds, and standard deviations (SDs) were reported to assess stability across splits. The approach balances thorough validation with computational efficiency within the TRE environment, providing multiple independent estimates of model performance.

3.5 Test Data Characteristics

The dataset consisted of paired radiology reports and CV-derived confidence scores from CXR images. Text inputs were encoded using 768-dimensional ClinicalBERT embeddings, while target outputs were continuous confidence scores (0-1) generated by the CV model.

4. Summary of Results

4.1 Performance Metrics & Key Findings

The linear regression model achieved an MAE of 0.061 ± 0.003 with an R^2 score of -2.325 ± 0.083, while the random forest model demonstrated slightly improved performance, producing an MAE of 0.033 ± 0.001 with an R^2 score of -0.610 ± 0.034 (mean ± SD across 5-fold cross-validation).

Classification metrics showed F-scores and Recall values of 0.000 ± 0.000 for both models, with 100% of predictions classified as “HARD” across all folds. Both models also produced a Balanced Accuracy of 1.000 ± 0.000 and MCC of 0.000 ± 0.000.

Table 1: Summary of regression metrics

Model	**MAE**	**R^2**
Linear Regression	0.061	-2.325
Random Forest	0.033	-0.610

4.2 Interpretation

Although the random forest model reduced prediction error, both models produced negative R^2 values, indicating performance worse than a baseline mean prediction. The narrow confidence intervals around the negative R^2 values suggest consistently poor performance across all validation folds rather than variance due to data splitting. Negative R^2 values indicate that the extracted textual features did not capture meaningful relationships linked to image complexity or model confidence.

Thresholding predicted confidence scores at 0.80 resulted in 100% of cases being classified as “HARD” under both models. This severe class imbalance indicates that the models were unable to generate predictions above 0.80. Consequently, binary classification accuracy could not be meaningfully assessed, and the aim of achieving the target 70% accuracy was unmet.

Both F1-score and Recall remained at 0.000 across all folds, demonstrating a consistent inability to identify non-“HARD” cases. Although a Balanced Accuracy of 1.000 appears to indicate perfect performance, this is an artefact of the dataset containing exclusively “HARD” cases. With no “EASY” cases present, the metric reduces to the recall of the single class and is therefore uninformative. The MCC score of zero further confirms the absence of any true discriminative capability.

4.3 Implications

The text-only prediction approach produced insufficient performance and would not be safe or reliable for identifying ambiguous cases in clinical practice. However, developing robust systems capable of prioritising challenging cases for expert review remains important for improving radiology workflows and supporting safe AI integration in healthcare.

The results suggest that difficulty prediction is likely a multimodal issue. Radiology reports primarily describe clinical findings rather than image-level characteristics that influence the confidence of CV models during classification. Future work could therefore combine text embeddings with image quality metrics, allowing prediction models to leverage both clinical text and image properties to link to CV model confidence, rather than text alone.

4.4 Conclusion

The results suggest that free-text in radiology reports does not contain meaningful linguistic signals for CV models' confidence scores for NGT placement detection. Linear regression and random forest models were unable to discern signals in the ClinicalBERT embeddings; the 100% “HARD” cases classified reflect this lack of predictive signals from report text.

4.5 Limitations

A key limitation of the project was that index-based matching was used instead of patient-level ID matching due to incompatible patient identifiers between datasets. Images and reports in the training data might not correspond to the same patient, which is a significant risk that would likely corrupt the label pairs the model learned from. If the difficulty labels and text reports were not paired properly, then the negative interpretation of results may not be indicative of failure stemming from the research question.

Another limitation was the reliance on confidence scores generated by a single ResNet-50 model to create ground-truth difficulty labels. As a result, the labels reflected the behaviours and biases of one specific architecture, limiting the generalisability of the findings. For example, cases labelled “HARD” may have been interpreted differently by other classifiers. Future work might include validation using an NGT-specific model to produce more representative ground-truth labels.

Furthermore, the computational constraints within the TRE limited the use to a subsample of the dataset, weakening statistical power and the model's ability to identify linguistic signals. Using the whole dataset of approximately 19,000 reports with GPU optimisations could potentially allow for better results.

5. Conclusions & Key Lessons

This study investigated whether clinical radiology reports contain meaningful linguistic signals predictive of CV model confidence in NGT placement detection.

Results suggest that report text showed no detectable predictive relationship with CV model confidence. Both linear regression ($R^2 = -2.325$) and random forest ($R^2 = -0.610$) performed worse than baseline, with all predictions classified as “HARD”, indicating that radiology reports lack linguistic features predictive of image classification difficulty. These findings also highlight the inherent complexity of difficulty prediction and the limitations of unimodal text-based approaches.

Future work should explore multi-class difficulty classification (EASY/MODERATE/HARD) with improved confidence calibration, fine-tuning of ClinicalBERT on domain-specific radiology language, and extended validation to other imaging tasks and modalities.

Given ongoing radiology workforce pressures and the safety implications of NGT placement, developing reliable difficulty-prediction systems remains a clinically important goal that warrants continued investigation.

References

Boehringer, A.S., Sanaat, A., Arabi, H. and Zaidi, H. (2023). An active learning approach to train a deep learning algorithm for tumor segmentation from brain MR images. *Insights into Imaging*, [online] 14(1). doi:https://doi.org/10.1186/s13244-023-01487-6.

Creeden, A., McFadden, S., Rainey, C., Campbell, S., Ather, S., Hajilou, A., Bond, R., McAllister, P. and Woznitza, N. (2025). Accuracy of interpretation of nasogastric tube position on chest radiographs by diagnostic radiographers: A multi-case, multi-reader study. *Radiography*, [online] 31(1), pp.83–88. doi:https://doi.org/10.1016/j.radi.2024.10.022.

Drozdov, Ignat, Dixon, R., Szubert, B., Dunn, J., Green, D., Hall, N., Shirandami, A., Rosas, S., Grech, R., Puttagunta, S., Hall, M. and Lowe, D.J. (2023). An Artificial Neural Network for Nasogastric Tube Position Decision Support. *Radiology: Artificial Intelligence*. doi:https://doi.org/10.1148/ryai.220165.

He, K., Zhang, X., Ren, S. and Sun, J. (2016). Deep Residual Learning for Image Recognition. *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, [online] pp.770–778. doi:https://doi.org/10.1109/cvpr.2016.90.

Huang, K., Altosaar, J. and Ranganath, R. (2019). ClinicalBERT: Modeling Clinical Notes and Predicting Hospital Readmission. *arXiv:1904.05342 [cs]*. [online] Available at: https://arxiv.org/abs/1904.05342.

Javaid, M., Haleem, A., Ravi Pratap Singh and Ahmed, M. (2024). Computer Vision to Enhance Healthcare Domain: An Overview of Features, Implementation, and Opportunities. *Intelligent pharmacy*, 2(6). doi:https://doi.org/10.1016/j.ipha.2024.05.007.

Johnson, A.E.W., Pollard, T.J., Berkowitz, S.J., Greenbaum, N.R., Lungren, M.P., Deng, C., Mark, R.G. and Horng, S. (2019). MIMIC-CXR, a de-identified publicly available database of chest radiographs with free-text reports. *Scientific Data*, [online] 6(1), p.317. doi:https://doi.org/10.1038/s41597-019-0322-0.

Lindroth, H., Nalaie, K., Raghu, R., Ayala, I.N., Busch, C., Bhattacharyya, A., Moreno Franco, P., Diedrich, D.A., Pickering, B.W. and Herasevich, V. (2024). Applied Artificial Intelligence in

Healthcare: A Review of Computer Vision Technology Application in Hospital Settings. *Journal of Imaging*, [online] 10(4), p.81. doi:https://doi.org/10.3390/jimaging10040081.

Liu, X., Faes, L., Kale, A.U., Wagner, S.K., Fu, D.J., Bruynseels, A., Mahendiran, T., Moraes, G., Shamdas, M., Kern, C., Ledsam, J.R., Schmid, M.K., Balaskas, K., Topol, E.J., Bachmann, L.M., Keane, P.A. and Denniston, A.K. (2019). A comparison of deep learning performance against health-care professionals in detecting diseases from medical imaging: a systematic review and meta-analysis. *The Lancet Digital Health*, 1(6), pp.e271–e297. doi:https://doi.org/10.1016/s2589-7500(19)30123-2.

The Royal College of Radiologists (2024). *Clinical Radiology Workforce Census 2024*. [online] Available at: https://www.rcr.ac.uk/media/4imb5jge/_rcr-2024-clinical-radiology-workforce-census-report.pdf.

Vermeulen, C., Pagès-Gallego, M., Kester, L., Mariëtte E.G. Kranendonk, Wesseling, P., Verburg, N., De, P.C., Evert Jan Kooi, L. Dankmeijer, Jasper, Kirsten van Baarsen, Hoving, E.W., Bastiaan B.J. Tops and Jeroen de Ridder (2023). Ultra-fast deep-learned CNS tumour classification during surgery. *Nature*, 622. doi:https://doi.org/10.1038/s41586-023-06615-2.

Xu, H., Maccari, B., Hervé Guillain, Herzen, J., Agri, F. and Jean Louis Raisaro (2023). An End-to-End Natural Language Processing Application for Prediction of Medical Case Coding Complexity: Algorithm Development and Validation. *JMIR medical informatics*, 11, pp.e38150–e38150. doi:https://doi.org/10.2196/38150.

Zech, J.R., Badgeley, M.A., Liu, M., Costa, A.B., Titano, J.J. and Oermann, E.K. (2018). Variable generalization performance of a deep learning model to detect pneumonia in chest radiographs: A cross-sectional study. *PLOS Medicine*, [online] 15(11), p.e1002683. doi:https://doi.org/10.1371/journal.pmed.1002683.